



Outbreak Risk Horizon Scan — 2026

A SENTINEL- Φ early-warning assessment of global viral spillover & amplification risk · InfecToNET, London School of Hygiene & Tropical Medicine · infec-tonet.org

Executive summary

InfecToNET's pilot early-warning engine, **SENTINEL- Φ** , fuses three orthogonal signal classes — **ecological forcing** (ENSO/ONI, NDVI, rainfall, soil moisture, Indian Ocean Dipole, deforestation, conflict), **genomic amplification** (recent sequencing/lineage growth), and **event corroboration** (WHO/PAHO/WOAH alerts) — into a single 0–100 Spillover/Amplification Index (SAI) per pathogen, with tiered WATCH/ADVISORY/WARNING alerts. This brief summarises the current global risk horizon from a live SENTINEL- Φ sweep and recommends risk-reduction actions.

Key message: the most actionable signal is genomic amplification (G). Four pathogens show a real recent genomic surge alongside ecological pressure — **avian influenza H5**, **chikungunya**, **Oropouche**, and **Crimean-Congo haemorrhagic fever** — and warrant priority preparedness now.

Priority pathogens (live SENTINEL- Φ sweep)

Pathogen	SAI	Tier	E·G·N	Scenario & region of concern
Avian influenza H5	86	WARNING	72·100·100	Global/pandemic. H5N1/H5N5 2.3.4.4b spilling to mammals (US cattle, polar bear & walrus Svalbard, Antarctica). Watch mammalian-adaptation mutations.
Chikungunya	86	WARNING	72·100·100	Asia · Americas · S. Europe. Aedes albopictus expanding to temperate zones (E1-A226V).
Oropouche	82	WARNING	72·100·80	Amazon → Caribbean. Unprecedented 2024 surge; Cuba first detection; vertical-transmission concern.
Crimean-Congo HF	79	ADVISORY	58·100·100	Turkey/C. Asia → Iberia. Hyalomma tick moving north with warming; autochthonous cases in Spain.
Mpox (clade I/IIb)	69	WARNING	75·38·100	Central/East Africa. Clade IIb cross-border spread from DRC.
Ebola / Marburg	68/64	WARNING	96·0·100	DRC / Albertine Rift. Bundibugyo PHEIC active; amplified by M23 conflict & health-system collapse.
Dengue / Yellow fever	61/63	WARNING	82·0·100	Brazil hyperendemic. El Niño + high NDVI + saturated soil drive Aedes/Haemagogus.
Hantavirus (ANDV)	56	WARNING	72·0·100	Patagonia. 2023-24 El Niño lag; 2025-26 cruise-ship outbreak; person-to-person ANDV.

E = ecological forcing · G = genomic amplification · N = event corroboration (each 0–100). Source: live /api/earlywarning sweep, InfecToNET, June 2026.

Driving mechanisms (from InfecToNET EcoRisk signals)

- **El Niño 2023–24 (ONI +2.06, 5th strongest since 1950)** → Patagonia hantavirus (12–18-mo lag); Amazon dengue/Oropouche/yellow fever (rainfall + NDVI + saturated soil); Horn of Africa RVF if the Indian Ocean Dipole turns positive.
- **Warming + Aedes/Hyalomma range expansion** → chikungunya & dengue into southern Europe and the temperate USA; CCHF tick into Iberia and central Europe.
- **Migratory bird flyways + H5** → HPAI already at both poles (Arctic Svalbard, Antarctica); next threshold is sustained mammal-to-mammal adaptation.
- **Armed conflict in eastern DRC (M23)** → Ebola, Marburg and mpox amplified as surveillance and contact-tracing collapse.

Recommended risk-reduction actions

- 1. Climate-indexed pre-season alerts.** Trigger SENTINEL-Φ WATCH alerts when $ONI \geq +1.5$: enhanced rodent surveillance in Patagonia and vector control in the Amazon / Horn of Africa **up to 12 months ahead**.
- 2. One Health animal–human interface.** Sequence H5 in mammals (cattle, seals, bears) to catch adaptation mutations (e.g. PB2 D701N); monitor fruit-bat reservoirs in the Albertine Rift and Congo Basin.
- 3. Decentralised genomic capacity.** The genomic channel only fires where sequencing happens. Close the critical gaps — Brazil (dengue under-deposited) and conflict-affected Central Africa — with pre-positioned portable sequencing.
- 4. Conflict-sensitive response.** For Ebola/Marburg/mpox in the DRC, establish humanitarian surveillance corridors; conflict is the dominant structural amplifier.
- 5. Temperate vector vigilance.** Entomological surveillance of *Aedes albopictus* and *Hyalomma* ticks at the new climatic frontiers in southern Europe and the USA.

What InfectoNET uniquely enables: a single open, cryptographically-verifiable score that correlates the ecological *driver* + the genomic *response* + news *corroboration* — moving surveillance from reactive to anticipatory, with 2-week to 12-month lead depending on the pathogen.

SENTINEL-Φ is a research pilot and not a substitute for official public-health alerts from WHO, PAHO or national authorities. Data: NCBI, GISAID, Nextstrain, NOAA, Open-Meteo, NASA MODIS, World Bank, ACLED. Contact: infectonet@gmail.com.